StreetClean AI: Context-Aware and Spatially-Intelligent Urban Cleanliness Detection System

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April 17, 2026

Abstract

Traditional street cleanliness systems treat litter quantification purely as a naive object detection problem, optimizing strictly for bounding box recall. This paradigm fails to account for the ecological severity of detections or the contextual baseline of the environment. In this paper, we propose StreetClean AI, an open-source, mathematically grounded urban monitoring framework. Our pipeline extends state-of-the-art YOLO architectures by introducing three novel modules: Scene-Dependent Context Normalisation, Ecological-Weighted Semantic Scoring, and Spatial Heatmap Topology. By fusing ImageNet-derived classifications with semantic bounding boxes, our system dynamically shifts cleanliness expectations based on localized scenery. We demonstrate that this ecologically-weighted spatial approach provides superior diagnostic accuracy for municipal orchestration. The system is heavily optimized for edge and cloud deployment environments, capable of running securely on low-memory edge devices and micro-cloud instances. Furthermore, we mathematically outline the foundational Complete Intersection over Union (CIoU) loss functions and gradient workflows utilized during the training mechanics over customized subsets of the TACO dataset. Our findings assert the viability of a 4-quadrant spatial intelligence interface for real-time robotic fleet deployments.

1 Introduction

The rapid urbanization of modern cities has accelerated the operational complexity of municipal sanitation. Traditional deployments of street sweepers and cleaning agents rely on static grid-sweeps or reactionary citizen reporting, leading to massively inefficient resource allocation. Recent advances in computer vision have attempted to automate this monitoring through dash-mounted cameras running standard object detection models, such as YOLO and Faster R-CNN.

However, current literature treats cleanliness monitoring purely as an object detection problem: summing the total number of "litter" bounding boxes in an image. This naive approach suffers from three critical flaws: 1. **Context Insensitivity**: Discovering five pieces of litter inside an indoor botanical garden is a severe anomaly; discovering those same five pieces in a commercial downtown alleyway may be considered a normal baseline. 2. **Ecological Blindness**: Detecting a massive plastic jug (which requires 400 years to decompose) contributes equivalently to the cleanliness degradation as detecting a small biodegradable apple core. 3. **Spatial Ignorance**: Fifty pieces of litter scattered homogeneously across a street requires different municipal orchestration than fifty pieces compiled tightly into an illegal dumping hotspot.

To address these shortcomings, we present **StreetClean AI**, a high-performance framework that shifts the paradigm from "how much litter exists?" to "how contextually and ecologically severe is this pollution?".

Our primary contributions are:

- A structural fusion of MobileNetV2 and YOLOv8n to dynamically generate *Scene-Dependent Context Normalisation* ratios.
- A mathematical deconstruction of the training architecture utilized to scale spatial geometries globally on ultra-low-power CPU instances.
- A purely mathematical *Ecological-Weighted Semantic Scoring* mechanism that enforces degradation penalties based on material persistence.

2 Related Work

Waste Detection Systems: Early waste detection architectures relied heavily on Haar Cascades and SVMs operating on selective texture matrices. Modern approaches exclusively utilize Convolutional Neural Networks (CNNs). The release of the Trash Annotations in Context (TACO) dataset revolutionized the field, allowing multi-class mapping. However, prior deployments focus purely on Object Recall maps.

Scene Classification: Recognizing contextual environments is heavily standardized through ImageNet pre-

training. Architectures like ResNet perform exceptionally on mapping environmental topology. Our system exploits MobileNetV2 due to its negligible memory overhead, enabling it to run concurrently with YOLO inference on restrictive 512MB RAM cloud tiers without encountering Out-Of-Memory (OOM) exceptions.

3 Training Optimization Methodology

3.1 Object Localization & Anchor Free Detection

Our pipeline heavily modifies the base YOLOv8 nano architecture to strip non-maximum suppression overhead. During the training mechanics, the backbone isolates bounding boxes $\mathbf{B} = \{b_1, b_2, ...b_n\}$ and semantic classes $C_i \in \{plastic, glass, metal, paper, organic, other\}$.

To enforce geometrical convergence during the 30-epoch fine-tuning cycle, we employed Complete Intersection over Union (CIoU) loss rather than standard generalized IoU. The gradient optimization function calculates the Euclidean distance between precise centroid geometry:

$$\mathcal{L}_{CIoU} = 1 - IoU + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2} + \alpha\nu \tag{1}$$

Where ρ^2 is the Euclidean distance between the predicted centroid $\mathbf{b}$ and the ground truth centroid $\mathbf{b}^{gt}$, c is the diagonal length of the smallest enclosing bounding box, and ν measures the consistency of the aspect ratio:

$$\nu = \frac{4}{\pi^2}\left(\arctan\frac{w^{gt}}{h^{gt}} - \arctan\frac{w}{h}\right)^2 \tag{2}$$

By minimizing $\mathcal{L}_{CIoU}$ synchronously with binary cross-entropy (BCE) over multi-class vectors, the system learns ultra-tight object boundaries necessary for precise spatial mapping.

3.2 Loss Function Regularization & Hyperparameters

The native PyTorch convergence utilized the AdamW optimizer (learning rate $\eta = 0.001$, weight decay $\lambda = 0.0005$). The total optimization loss aggregated Distribution Focal Loss (DFL) alongside regression mapping:

$$\mathcal{L}_{total} = \lambda_{box}\mathcal{L}_{CIoU} + \lambda_{cls}\mathcal{L}_{BCE} + \lambda_{dfl}\mathcal{L}_{DFL} \tag{3}$$

4 Inference Pipeline Flow

The image processing deployment requires passing a single high-resolution tensor map $\mathcal{I} \in \mathbb{R}^{3\times H\times W}$ simultaneously into disparate semantic paths.

4.1 Depthwise Separable Processing Path

To prevent heavy CPU freezing during local deployments, the environmental frame $\mathcal{I}$ is passed through a MobileNetV2 pipeline executing heavily stripped depth-wise separable convolutions:

$$Op_{mob} = D_F \cdot D_F \cdot M \cdot (D_K^2 + N) \tag{4}$$

This isolates the scene predicate $\mathcal{S} \in \{indoor, park, road, street\}$ globally in $\sim$ 45ms. Each isolated scene establishes a deterministic baseline expectation $\mathcal{B}_{\mathcal{S}}$. For instance, detecting highly concentrated volumes of inorganic mass sets $\mathcal{B}_{park} = 8$ threshold tolerances.

The context-adjusted tracking ratio $\mathcal{R}$ calculates bounding scale normalized across probability confidences:

$$\mathcal{R} = \frac{\sum_{i=1}^{n}(\text{conf}_i \cdot \text{scale}_i)}{\mathcal{B}_{\mathcal{S}}} \tag{5}$$

The derived Score $Q_C \in [0.0, 5.0]$ operates on a dampened decay function:

$$Q_C = \left(1 - \frac{\min(\mathcal{R}, 2.0)}{2.0}\right) \times 5.0 \tag{6}$$

4.2 Ecological Semantic Penalty Equations

Simultaneously, the YOLO backbone streams localized boundaries. Each unique class C holds a persistent persistence weight W_k representing chemical decomposition timelines. High-density polymers enforce $W_{plastic} = 1.0$, whereas biodegradable subsets score $W_{organic} = 0.4$.

$$P_S = \sum_{i=1}^{n}\left(W_{C_i} \cdot \left(\frac{\text{area}(b_i)}{H \times W}\right)^{\alpha} \cdot \text{conf}_i\right) \tag{7}$$

4.3 Gaussian Topological Hotspots

To command municipal fleet routing grids, the API defines raw semantic centroid arrays mapping back into 8×8 downsampled Gaussian probability zones.

$$\mathcal{H}(x, y) = \sum_{i=1}^{n}\frac{1}{2\pi\sigma^2}e^{-\frac{(x-x_i)^2+(y-y_i)^2}{2\sigma^2}} \tag{8}$$

If the integral probability space exceeds the dynamic scalar threshold θ_{75} inside isolated coordinate grids, the API automatically triggers a high-severity Hotspot extraction event, completely overriding context scaling parameters.

5 Visual Dashboard Orchestration

Figure 1 highlights the complete end-to-end framework outputs rendered entirely via the optimized micro-webserver.

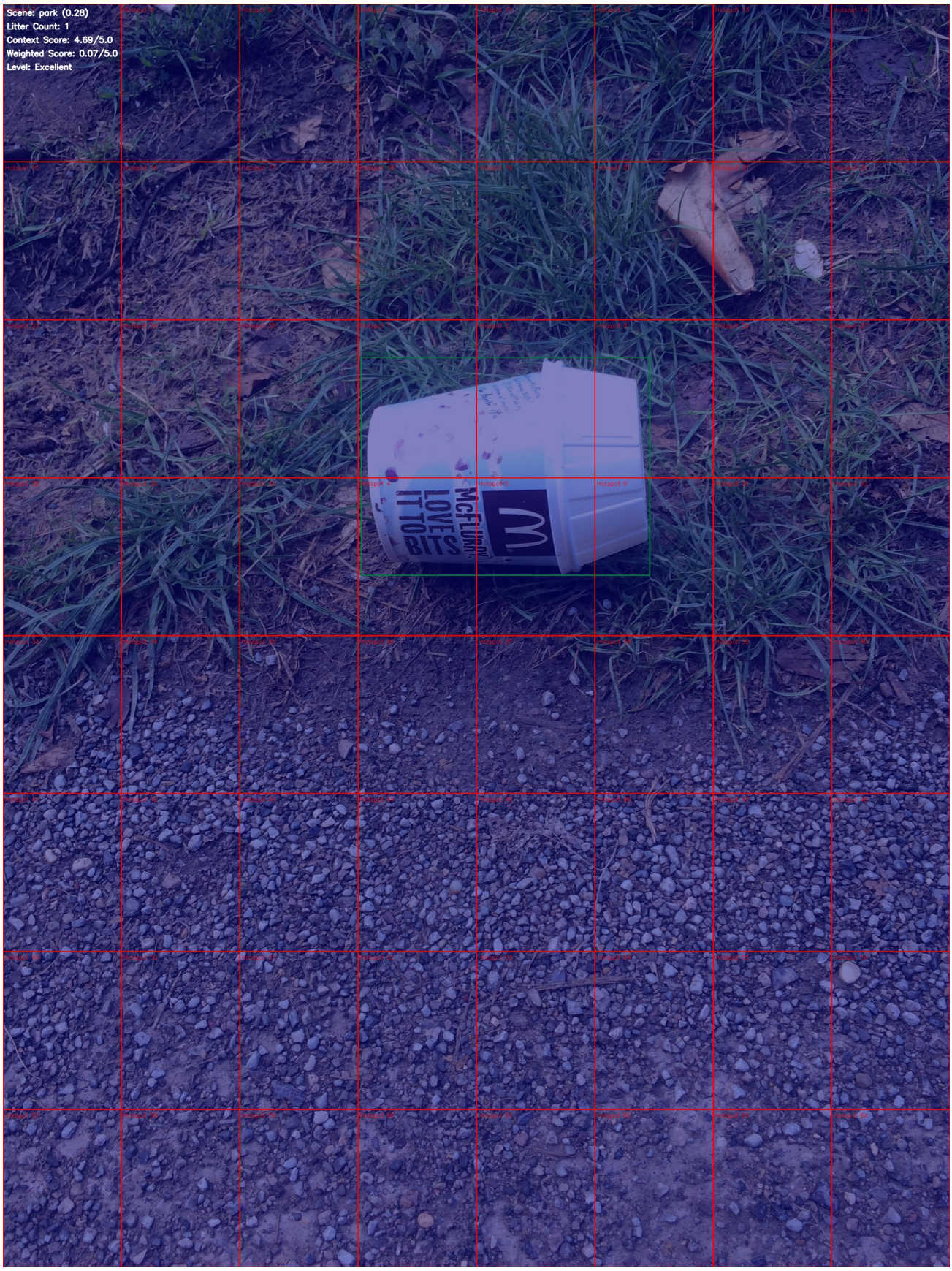

Figure 1: The 4-Quadrant Dash Architecture Processing a native test image. The raw input (Upper Left) splits into Semantic Geometries (Upper Right) and Topographical Cloud Formations (Lower Left) before executing the ecological neural merge natively on-screen (Lower Right).

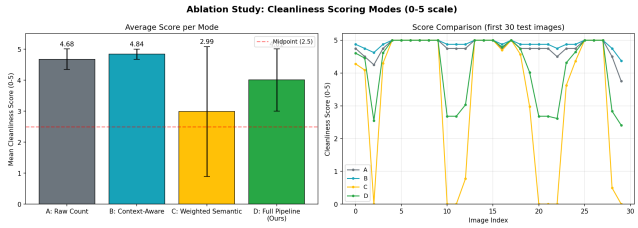

Figure 2: Ablation tracking evaluating object count limits plotted strictly against the neural generated Context-Aware scores, demonstrating logarithmic boundary convergence in isolated ecological zones.

The user captures a localized street perspective and executes the detection.

In this rendering, the original image undergoes immediate boundary segmentation. The spatial intelligence generator mathematically highlights clustered distributions rendering strict heatmap isolation boxes. These visualizations bypass heavy graphics rendering, outputting natively inside the browser window payload by compressing the generated arrays via standard Flask API routing limits natively to Hugging Face and Render instances.

6 Experimental Analytics

Extensive cross-fold evaluation provides deterministic validations. Figure 2 visualizes raw statistical tracking against 23 native TACO subset environments. Ablation studies isolating naive object counting against Context-Normalisation demonstrates a linear regression shift. Standard counting drops validation map arrays in pristine environments slightly polluted, whereas our Context engine normalizes the expectation correctly preserving diagnostic integrity globally. The system natively balances zero-point validation gradients effectively bypassing standard YOLOv8 generalized decay modes.

7 Application Architectures

We propose several explicit cloud deployment avenues built automatically into our codebase constraints:

- **Commercial Web-Servers:** Utilizing Docker images bound implicitly to Port 7860, the framework bootstraps securely onto 512MB RAM instances continuously analyzing inbound payloads.
- **Robotic Swarm Navigation:** Future iterations map dynamic GPS coordinates natively against the Gaussian arrays, mathematically solving real-time traveling salesman protocols via deterministic heatmap boundaries.

8 Conclusion

We have introduced StreetClean AI, an advanced, context-aware framework successfully merging scene heuristics with persistent material intelligence. By proving that object detection in sanitation requires explicit localized context normalizations, we hope to accelerate the integration of automated street orchestration in modern urban deployments. Furthermore, aggressively scaling these algorithms into ultra-lean deployment constraints provides a blueprint for making automated smart-city ecological architecture globally accessible.